What Do You Mean? Exploring the Alleged Theory of Mind Abilities of Large Language Models

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Abstract

This study explores the capacity of Large Language Models (LLMs) to perform tasks requiring Theory of Mind (ToM), a critical component of pragmatic language understanding. Although previous work suggests that LLMs may exhibit emergent ToM abilities, this research examines whether such capabilities genuinely involve reasoning about beliefs or merely reflect the reliance on shallow statistical cues. Through a series of controlled experiments featuring indirect speech acts and verbal irony, we assess how belief contexts influence LLM interpretations. The results reveal that, although LLMs occasionally succeed in decoding communicative intentions, their performance is not attributable to genuine ToM reasoning. Instead, models often depend on conventionalized linguistic cues rather than engaging in true mentalizing. This work underscores the limitations of LLMs in simulating human-like ToM and offers insight into their interpretive biases, contributing to a deeper understanding of their linguistic and cognitive capabilities [1].

1 Introduction

An hallmark of human communication is the pragmatic ability to reconstruct the intended meaning of an utterance (that is, what the speaker intends to convey with it in a given situation), even when it goes beyond its literal interpretation (Grice, 1975; Stephen C. Levinson, 2000; Sperber and Wilson, 2002; Pinker et al., 2008). In this context, the **Theory of Mind** (ToM) – the ability to attribute mental states to others – is generally regarded to be essential for understanding intended meaning and crucial for successful social interactions in everyday life (de Villiers, 2007).

[1]The dataset and code will be made publicly available on GitHub.

Recent advancements in Large Language Models (LLMs) and the unforeseen emergence of new capabilities in tasks for which these models were neither explicitly trained nor instructed have prompted increased research into whether ToM could be among these newly emerging skills (Marchetti et al., 2023; Kosinski, 2023).

Determining whether these models can handle tasks that require the use of ToM would enable us to: 1) revise our understanding of ToM development from a linguistic or psycholinguistic perspective, 2) assess if the language generated by models can be compared to human language and whether they use the same cues to perform tasks and produce texts, and 3) evaluate their effectiveness as communicators.

Kosinski (2023) claim that ToM can be regarded as one of the **emerging abilities** of large language models (LLMs). However, many of these studies overlook the conventionalization of expressions and the linguistic context, which are well-encoded in the vast amounts of text acquired during pretraining (Lu et al., 2024).

Many works in psycholinguistics (Gibbs, 1983, 2002) argue that in the presence of utterances with non-literal meanings, certain cues in the situational and linguistic context can directly lead to a non-literal interpretation, bypassing the literal interpretation stage. Thus, non-literal interpretations often become the default, particularly with conventionalized or idiomatic expressions. This applies to phenomena requiring mentalizing, such as **Indirect Speech Acts** (ISAs) (Gibbs, 1986b; Marocchini and Domaneschi, 2022) and **Verbal Ironies** (Ackerman, 1983; Juez, 1998; Burnett, 2015).

Building on this evidence, we hypothesize that, similar to humans, conventionalization in LLMs can play a role in enabling the completion of tasks typically considered to require mentalizing, without the need of accounting for ToM.

We aim to understand the following questions: 1) To what extent do LLMs accurately perform tasks that require mentalizing? 2) If LLMs exhibit such capabilities, can their performance be shaped or enhanced by the conventionalization of linguistic expressions? 3) How do the error patterns of LLMs compare to those of humans in terms of informativeness? 4) Does the size of the language model significantly influence its accuracy in mentalizing tasks? 5) Is the performance of LLMs in these tasks indicative of their ability to simulate ToM and account for varying belief states within communicative contexts?

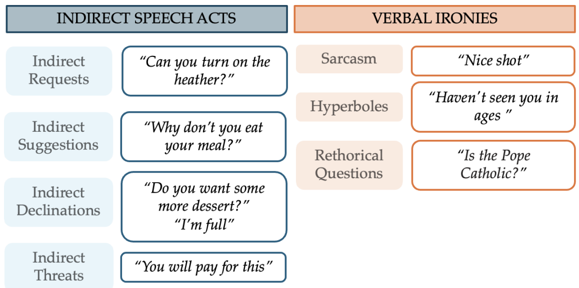

Figure 1: Linguistic phenomena included in our dataset: Indirect Speech Acts and Verbal Ironies

2 Related Work

Mahowald et al. (2024) argue that models employing language in human-like ways, such as LLM, must develop two distinct competencies: **formal linguistic competence** – knowledge of linguistic rules and patterns – and **functional linguistic competence** –understanding and using language in real world contexts. These competencies require the development of specialized mechanisms. This highlights the need for separate testing and evaluation of these competencies through benchmarking techniques and stimuli that isolate either of them.

Our research aligns with the perspective of Hu et al. (2023), who compare language models and humans across seven pragmatic phenomena using zero-shot prompting to evaluate the models' ability to interpret non-literal language. In contrast, our study diverges in its focus on pragmatic phenomena that require mentalizing abilities. Specifically, we seek to determine whether LLMs interpret non-literal language by inferring others' beliefs, or rely on other conventionalized cues to derive the intended meaning. Existing benchmarks for assessing ToM in models have predominantly addressed social commonsense (Sap et al., 2019), situational descriptions (Nematzadeh et al., 2018; Le et al., 2019; Sap et al., 2022; Zhou et al., 2023), or social interactions (Kim et al., 2023; Li et al., 2023).

Most studies in ToM primarily investigate false belief tasks (Baron-Cohen et al., 1985), which assess whether a model can represent the beliefs of another agent that are factually incorrect but consistent with the observations of that agent (Jones et al., 2023; Shapira et al., 2023). These works focus on first or second-order ToM, while Hu et al. (2023) investigates third-order and beyond.

LMs have been shown to succeed on some ToM tests (Kosinski, 2023) while failing on others (Gandhi et al., 2023; Chen et al., 2024; Ullman, 2023). As suggested in Ma et al. (2023b), these varied results can be attributed both to differences in evaluation methods and to the variation in the types of prompt used in the studies, as occurs in (Strachan et al., 2024). Moreover, many current findings can be explained through shortcuts, arising because the tasks used to investigate ToM in deep learning systems have been too narrow (Aru et al., 2023). To the best of our knowledge, the present study is the first to control the role of different belief contexts in tasks requiring ToM reasoning.

3 Experimental Materials

3.1 Linguistic phenomena

It is a common assumption that Indirect Speech Acts and Verbal Ironies require mentalizing for their highly context-dependent nature.

Indirect Speech Acts (ISAs) (Austin, 1975; Searle, 1975) occur when the speaker's intended meaning differs from the literal meaning of their words. They rely on the listener's ability to infer intent from context and cues, often used to achieve politeness or convey subtle meanings indirectly.

We assessed four distinct types of ISAs, each differing in their level of conventionalization: **Indirect Requests** (Trott and Bergen, 2019), **Indirect Suggestions**, **Indirect Declinations**, and **Indirect Threats** (Figure 1). Accurately interpreting these ISAs may not solely depend on the ability to encode and represent others' knowledge states and intentions; it could also arise from conventional associations. In our experiments, we controlled

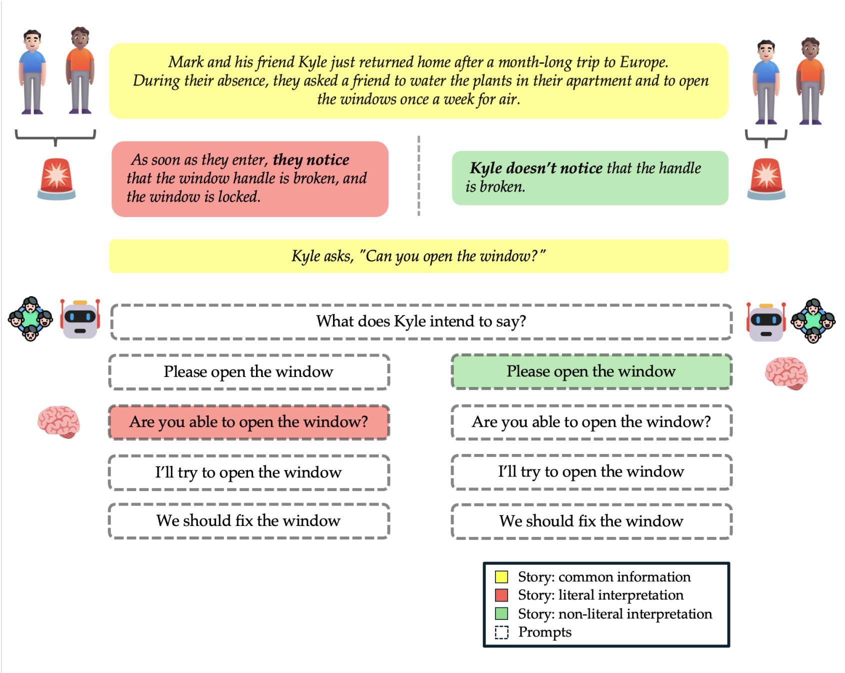

Figure 2: Stimulus from Indirect Requests illustrates two interpretations of the story: non-literal (in green) and literal (in red). Each story involves two characters whose knowledge of the situation varies between the two versions. The yellow boxes highlight the common information retained across both versions. Both models and humans are presented with the same question and must select one of four possible answers.

for both literal and non-literal interpretations, so that differentiating between them necessitates considering ToM, as the relevant contexts differ based on the mutual beliefs held by the speaker and listener.

Verbal irony (Skalicky, 2024) encompasses utterances where the speaker feigns ignorance of an unfulfilled expectation, whether based on personal anticipations or cultural norms. This pretense creates a discrepancy between the spoken words (reflecting the expectation) and the actual situation.

According to the graded salience hypothesis (Giora and Fein, 1999a), both literal and figurative meanings of phrases can be encoded in the lexicon based on their conventionality.[2] For example, *Thanks a lot* is used both sincerely and ironically. Consequently, figurative meanings of conventional ironic expressions are more accessible and easier to comprehend than those of situation-specific ironic expressions, which may require additional contextual cues or processing. We investigated three kinds of verbal ironies: **Sarcasm** (Gibbs, 1986a), **Hyperboles** (Aguert et al., 2018; Carston and Wearing, 2015), **Rethorical Questions** (Figure 1)

[2] Ackerman (1983) suggested that some ironic remarks might have become idiomatic, as children interpreted them ironically with minimal contextual cues.

3.2 Stimuli

The stimuli used in our study were inspired by the False-Belief task design, and adapted from Trott and Bergen (2019), who created pragmatically ambiguous utterances to conclude stories. In these stories, an **obstacle** to fulfil a potential request was introduced, with the speaker either present or absent when the obstacle information was disclosed. We incorporated Trott and Bergen (2019)'s stimuli on Indirect Requests into our experiment, categorizing them into two groups: IR-Os (derived from Trott and Bergen (2019) IR-Ns (created by us). Stimuli for all other phenomena were newly developed to ensure they were not included in the pretraining material.

Each story in the dataset presents two interpretations (Figure 2): in one version the belief scenario is congruent with the literal interpretation preventing any other non-literal meaning, while in other version the belief context triggers an indirect interpretation. The concluding target utterances in each story are conventionalized to typically convey a non-literal meaning. For Verbal Irony, conventionalization is achieved through conventional remarks following story contexts with varying degrees of salience (Giora and Fein, 1999b): higher salience (e.g., a negative context followed by a positive outcome) or lower salience (e.g., a positive context aligned with a positive outcome).

This design enabled us to assess whether models adjusted their interpretations according to the different belief contexts of the two participants in the story. Successful task completion requires both humans and models to infer different mental states (i.e., beliefs, intentions and emotions) of the two characters. Indeed, the narrative remains consistent across both literal and non-literal versions, with the distinction based on the characters' knowledge of the obstacle. Thus, identifying the correct interpretation given a certain context requires accurate ToM reasoning. Consistent with Hu et al. (2023), we also developed reduced context versions of each story to assess whether inference relies on cues within the linguistic context.

The final dataset comprises 160 stories for Indirect Speech Acts, including 80 with reduced context, and 96 stories for Verbal Ironies, of which 46 have reduced context.

4 Experimental setup

4.1 Task

We assessed both humans and models using a multiple-choice question format to enable a direct comparison between the two groups and ensure exposure to the same stimuli with minimal variations in prompts. Each story was followed by a question regarding the intended meaning of the character's utterance, phrased as: "What does ⟨name⟩ intend to say?" Respondents selected from four options: the non-literal interpretation, the literal interpretation, and two semantically coherent but pragmatically inappropriate distractors.

4.2 Human data

We tested 360 participants using the Prolific platform. Indirect Speech Acts (ISAs) were categorized into eight groups, while Verbal Ironies were divided into four groups, with each group comprising 20-21 stories. The distribution of phenomena was randomized to ensure their equal representation and a balanced number of both indirect and direct interpretations across groups. Each group was tested with 30 speakers.

The participants were native English speakers, equally balanced in gender, from the USA and the UK. The test included three check questions, and participants were compensated only if they answered these questions correctly (Appendix A.

Model	Parameters
GPT-2	117M
T5 (large)	770M
Flan-T5 (XL)	3B
Tk Instruct	3B
Falcon / Instruct	7B
Llama 2 / Instruct	7B
GPT-3.5	175B (est.)
GPT-4	1.76T (est.)

Table 1: Models tested in our experiments.

4.3 Models

We evaluated ten models (Table 1), ranging from those with smaller to larger numbers of parameters, including models fine-tuned for specific tasks to better align their performance with human responses. Our selection included several models assessed in Hu et al. (2023): **GPT-2** (Radford et al., 2019) 117M parameters, **Tk-Instruct** (Wang et al., 2022) 3B parameters, and **Flan-T5** XL (Chung et al., 2022) 3B parameters. Additionally, we tested base models such as **T5 large** (Chung et al., 2022) 770M parameters, **Falcon** (Almazrouei et al., 2023) 7B parameters, and **Llama 2** (Touvron et al., 2023) 7B parameters, along with their instructed counterparts: **Falcon Instruct** (Almazrouei et al., 2023) 7B parameters, **Llama 2 Instruct** (Touvron et al., 2023) 7B parameters. Finally, we also tested two proprietary instructed models: **GPT-3.5** (OpenAI, 2023) 175B parameters (estimated) and **GPT-4** (OpenAI, 2024) 1.76T parameters (estimated).[3]

Each model received the same prompt (Appendix A)[4]. The prompt included detailed task instructions, a request to select one of the proposed answers by its corresponding number (1, 2, 3, or 4), the story with the question, the possible options, and the final statement, *"The correct answer is:"*. Each model was presented with the same stimulus four times, with the order of the answers randomized on each occasion. In evaluating LLMs using multiple-choice questions, altering the number of response options does not inherently enhance accuracy and confidence. However,

[3] All non-OpenAI models were accessed via Huggingface (Wolf et al., 2020), while OpenAI were accessed with API call on February 2024

[4] Llama 2 was prompted with the [INST] token before and after the instruction.

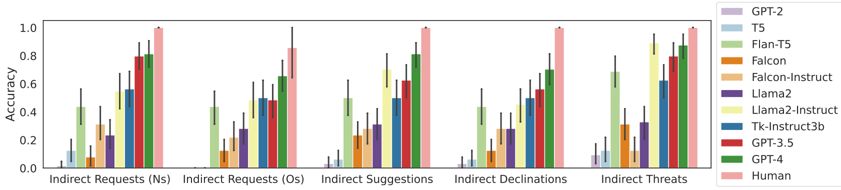

Figure 3: ISAs - Accuracy for tasks with non-reduced context.

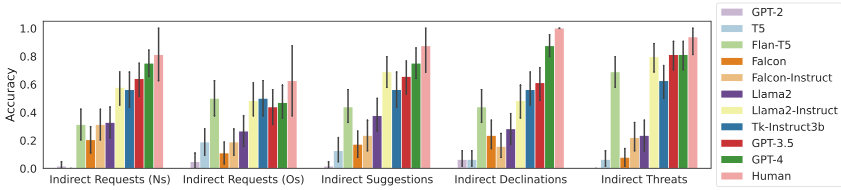

Figure 4: ISAs - Accuracy for tasks with reduced contexts.

rearranging the order of answers for each question and repeating questions can improve the robustness of the assessment process (Li et al., 2024).

4.4 Evaluation

Our evaluation paradigm employs zero-shot prompting to assess the knowledge acquired during training and investigate ToM as an emergent ability. We only considered model outputs that were numerical responses from 1 to 4; any other generated text was considered invalid and indicative of the model's failure to perform the task, resulting in a False label for such responses. An answer was deemed correct if the generated number matched the correct option.

5 Experimental Results

5.1 Indirect Speech Acts

Figure 3 displays the number of correct answers selected by both humans and models. Notably, models such as GPT-2, T5, Llama2, Falcon, and Falcon Instruct perform at chance levels across all phenomena.

The results suggest that instruction tuning has a greater impact on performance than model size alone. Models such as Flan-T5, Tk-Instruct-3B, and Llama2-Instruct exhibit superior performance compared to their non-instructed counterparts (T5 and Llama2). However, Falcon Instruct is not able to match the smaller instructed models. This variation may be due to differences in the instruction fine-tuning: Falcon Instruct uses a mix of instruct and chat datasets, Tk-Instruct incorporates diverse tasks including text categorization and dialogue generation, and Flan-T5 is fine-tuned with 473 datasets across 146 task categories. The synergy of instruction tuning and model size is evident in models like GPT-3.5 and GPT-4, which demonstrate the highest levels of performance. Nevertheless, Llama2-Instruct achieves notable success in indirect threats despite its smaller size.

It is worth noting that even the best models underperform with respect to humans across all tasks. The ISAs class with the lowest accuracy are Indirect Requests (Os), which show a similar performance pattern between humans and models, and Indirect Declinations. Tasks with reduced context reveal greater variability in human performance compared to models (Figure 4).

5.2 Irony

Figure 5 show that LLMS behave with ironies similarly to ISAs, with GPT-3.5 and GPT-4 achieving the highest performance levels. Instruction tuning appears to be more influential than model size in this context.

Reducted context for Indirect Hyperbole and Indirect Rethoric Questions show a severe decreasing in accuracy, mantaining however the model difference in accuracy registered for non reduced contexts (Figure 6).

6 Discussion and Analysis

In this section, we carry out a more comprehensive analysis of the experimental results to address the research questions outlined in Section 1.

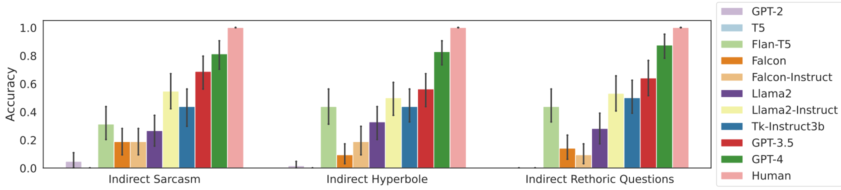

Figure 5: Irony - Accuracy for tasks with non-reduced context.

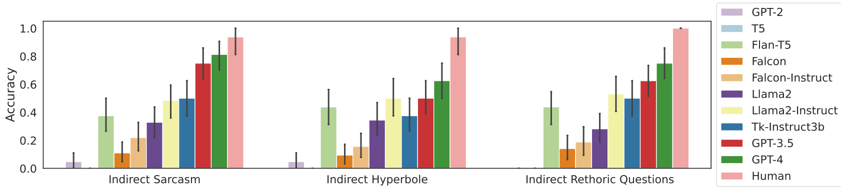

Figure 6: Irony - Accuracy for tasks with reduced contexts.

6.1 LLMs and Humans in mentalizing task

In our experiments, we controlled for both literal and non-literal interpretations, and differentiating between them necessitates considering ToM, as the relevant contexts differ based on the mutual beliefs held by the speaker and listener in the stories.

Can LLMs perform tasks that involve understanding mutual beliefs between characters? While LLMs do not achieve human-level accuracy in these tasks, certain models exhibit relatively high accuracy, particularly those that have undergone further training on specialized tasks. However, if one were to frame the question as requiring a binary answer, the response would be negative. The assessment of LLMs performance in mentalizing tasks should focus on the errors made rather than the number of correct answers. If LLMs genuinely possessed the ability to mentalize and apply such cognitive skills in task completion, their accuracy would likely approach that of humans. In humans, responses are consistently guided by their capacity for mentalizing, as ToM is either present or absent.

In this regard, our findings are consistent with numerous studies in the scientific literature (Ullman, 2023), which underscore the limitations of LLMs in executing tasks that necessitate mentalizing.

However, if ToM is either wholly present or absent, and we rule out the possibility of ToM-like capabilities in LLMs, what mechanisms or factors account for the accurate task completion observed in instances where correct answers are generated?

6.1.1 Models size vs. Accuracy

Instruction tuning appears to be more influential than model size, though combining instruction tuning with model scaling is particularly effective, as seen with models like GPT-3.5 and GPT-4. This effect is especially evident in tasks like ISAs, where instruction tuning improves performance. Models such as LLama2 Instruct and Tk-Instruct achieve accuracy levels comparable to GPT-4, particularly in tasks involving Indirect Threats. However, for tasks involving irony, model scaling alone leads to a more significant improvement in accuracy.

The higher accuracy in Indirect Threats is likely due to their frequent inclusion in instruction datasets, which helps align and fine-tune models. However, variability in correct answers across literal and non-literal contexts indicates that instruction tuning alone cannot fully explain some instances of accurate task completion.

6.1.2 Error Patterns

The data suggests that humans tend to struggle with literal interpretations when the most conventionalized interpretation is non-literal, as evidenced in tasks involving Indirect Requests (Os), where non-literal responses are more frequently chosen (Figure 7). Among the evaluated models, only GPT-4 exhibits an error pattern similar to humans in Indirect Requests, although it does not achieve human-level accuracy. For other ISAs, GPT-3.5 and GPT-4 generally prefer literal interpretations.

Proponents of the conventionalization hypoth-

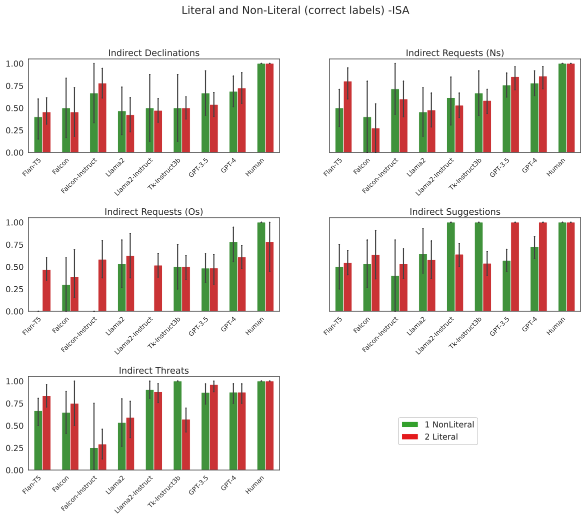

Figure 7: ISAs - Correct labels of Non-Literals and Literals

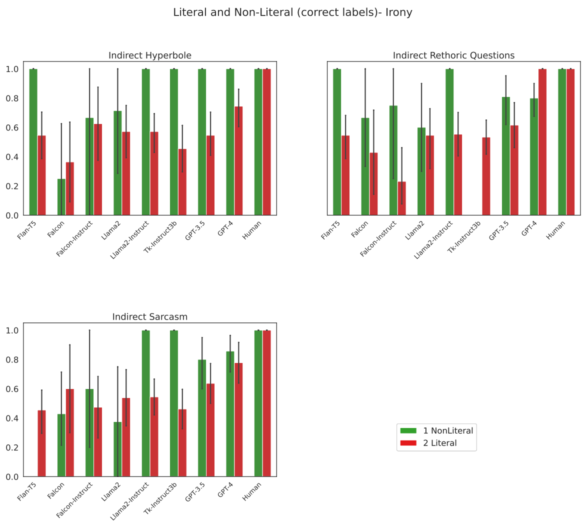

Figure 8: Irony - Correct labels of Non-Literals and Literals

esis (Gibbs, 1983, 2002) argue that humans typically select the most conventionalized reading, often the non-literal one in these tasks. However, none of the models consistently prefer the conventionalized reading over the literal one. Instead, they struggle equally with both interpretations, revealing their general difficulty in considering the belief context that would trigger or prevent the non-literal interpretation.

The case of irony is particularly noteworthy (Figure 8). Models with higher overall accuracy, such as GPT-4, GPT-3.5, LLama2-Instruct, and Tk-Instruct (with Flan-T5 performing well only in the context of Indirect Hyperbole), tend to struggle more with identifying the correct literal labels compared to non-literal ones in tasks involving Indirect Hyperbole and Indirect Sarcasm. In contrast, GPT-4 and Tk-Instruct tend to prefer literal interpretations in Indirect Rhetorical Questions, while other models favor non-literal interpretations.

Model	Hyperbole	Rhetoric Questions	Sarcasm
Flan-T5	†	†	-
Falcon	†	†	†
Falcon-Inst.	†	†	†
Llama2	†	†	†
GPT-3.5	†	*	*
GPT-4	*	*	†

Table 2: Irony (Non-literal) - Significance Comparison between ToM- without context and ToM°

This pattern suggests a correlation between higher accuracy and the model's tendency to perform better with non-literal interpretations, especially in irony. This could be influenced by the model's size and its exposure to data during the instruction phase. Models like GPT-3.5, GPT-4, LLama2-Instruct, and Tk-Instruct might be more sensitive to the conventionality of patterns and expressions present in their training or instruction datasets, leading them to rely on these conventionalized expressions for selecting the correct answer, rather than demonstrating a genuine ability to mentalize.

Furthermore, the similarity in error patterns observed in contexts with reduced information (Figure 4 and Figure 6), particularly in irony, suggests that context is not the primary factor in selecting the correct answer. Instead, it implies that the models may directly encode conventionalized expressions and their non-literal meanings, rather than relying on contextual cues to make the correct inference.

6.2 The role of conventionalization

Having excluded instruction tuning as the sole factor in accurate task completion and identified significant differences in processing patterns between LLMs and humans, further experimentation is needed to investigate the role of conventionalization in linguistic expressions. Our findings suggest that reducing contextual information does not result in different interpretations by the model. Therefore, we conduct additional experiments focused on modifying contextual features.

Model	Declinations	Requests (Ns)	Requests (Os)	Suggestions	Threats
Flan-T5	†	†	†	†	†
Falcon	†	†	†	†	†
Falcon-Instruct	†	†	†	†	†
Llama2	†	†	†	†	†
Llama2-Instruct	†	†	†	†	†
Tk-Instruct3b	†	†	†	†	†
GPT-3.5	†	†	†	†	†
GPT-4	*	†	†	†	†

Table 3: ISAs (Non-literal) - Significance Comparison between ToM- without context and ToM°

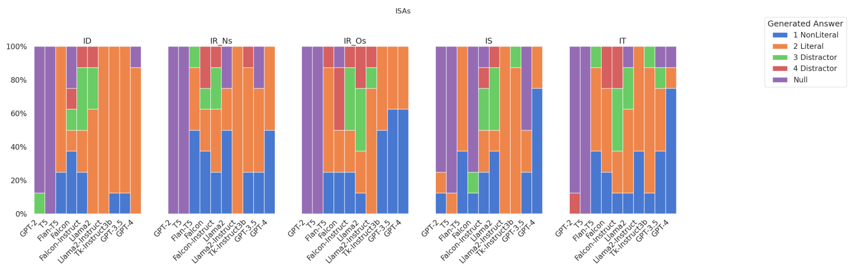

Figure 9: ISAs - Selected options for prompts ToM-

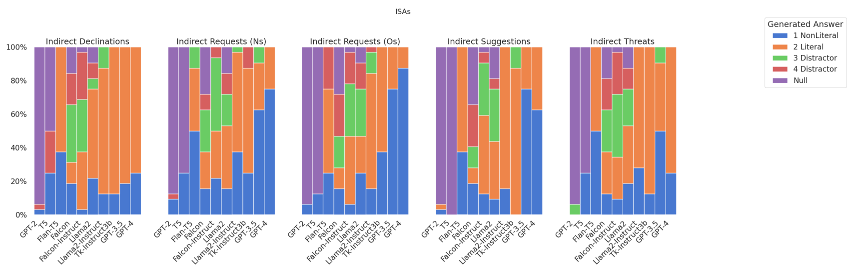

Figure 10: ISAs - Selected options for prompts ToM+

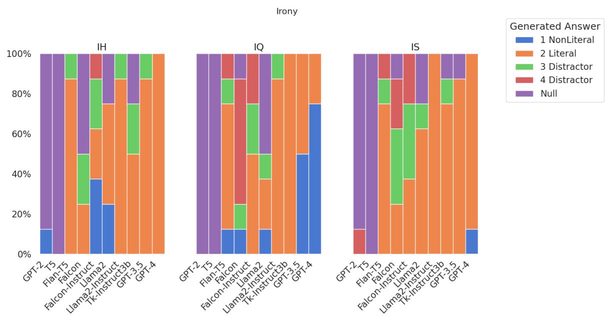

Figure 11: Irony - Selected options for prompts ToM-

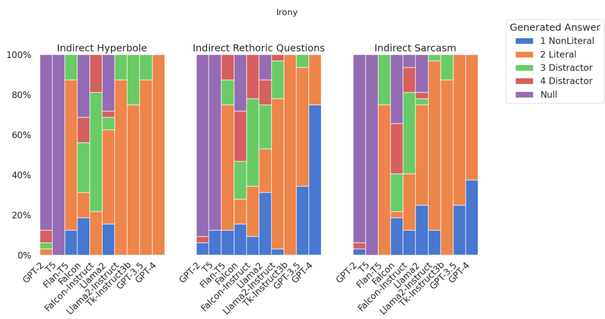

Figure 12: Irony - Selected options for prompts ToM+

6.2.1 Do belief contexts drive interpretation?

The additional experiments investigate whether the belief context in stories influences the model's ability to select the correct answer. We introduced two new prompt sets: (1) a set with only the target utterance, excluding contextual information **ToM-**, and (2) a set with a neutral context, omitting belief states that could prompt either a literal or non-literal interpretation **ToM+**. The original stimuli, which required ToM abilities, are referred to as **ToM°**.

Figures 9 and 11 demonstrate that GPT-3.5 and GPT-4 tend to favor non-literal interpretations of target sentences without context. This pattern holds even with neutral contexts (Figures 10 and 12), indicating that more accurate models may exhibit a bias toward non-literal meanings, following a gradient of conventionalization.

A detailed analysis of accuracy values, including chi-square statistics and Pearson residuals for both ToM- and ToM+ conditions, compared to ToM°, was conducted to assess whether observed differences between categories (phenomena and contexts) were statistically meaningful or likely to have occurred by chance. This analysis aimed to determine whether task completion and interpretations were systematically influenced by belief context across the phenomena under investigation.

In Tables 2 and 3 (see Appendix B for full data), we present the analysis of the significance of Pearson's residuals, using the following notations: ** for values greater than 4 ($\alpha \cong 0.0001$), * for values between 2 and 4 ($\alpha \cong 0.05$), and † for values below 2. As shown in the tables, only GPT-3.5 and GPT-4 exhibit any notable significance for irony, with GPT-4 also showing significance for ISAs in Indirect Declinations. In both instances, these phenomena were identified as the least conventionalized in our stimuli.

Moreover, our findings reveal that for most types of pragmatic phenomena there are no significant differences between ToM- and ToM°, even in the top models.

6.2.2 Belief or conventionalization?

The significance values underscore that less conventionalized phenomena are more likely to be influenced by context in models achieving higher accuracy, indicating that context plays a role in task completion. This correlation holds regardless of whether the items are presented in literal or non-literal form; less conventionalized items consistently rely more on context. In contrast, for other phenomena, models do not appear to depend on context when selecting an answer, opting for a response independently of the level of contextual detail provided. Moreover, there is no discernible difference between ToM- and ToM+ conditions, as the significance is moderate for both contexts in less conventionalized items, or not significant in either context.

Given these findings, what cues do the models use to select their answers?

If we consider the degree of conventionalization of different pragmatic constructions, a clear pattern emerges: less conventionalized Indirect Declinations result in the lowest accuracy, with Indirect Suggestions and Indirect Requests positioned in the mid-range, and the most conventional Indirect Threats (Holtgraves, 1991) yielding the highest accuracy (Figures 3 and 5).

In contrast, for irony-related tasks, the differences between the phenomena are less distinct. However, Indirect Rhetorical Questions seem to be the most conventional, followed by Indirect Sarcasm and Indirect Hyperboles.

Although no significant difference is observed between non-literal and literal stories, and conventionalization does not seem to aid models in correctly inferring beliefs, it nonetheless plays a role in the model's ability to rely on context. Unlike humans, the most conventional non-literal interpretation does not appear to be the default for the models. However, when examined in relation to context, conventionalization does appear to have an impact: the best-performing models (such as GPT-4) seem to rely more on context when dealing with less conventionalized phenomena, as evidenced by a significant difference in performance across different context patterns. In contrast, more conventionalized phenomena do not show any noticeable difference in performance, regardless of whether the context is rich or sparse.

7 Conclusions

Previous works have reported outstanding performances of LLMs in tasks that are typically assumed to require ToM. The aim of this paper was to investigate such alleged ToM abilities when LLMs are prompted to decode the intended meaning of ISA and Verbal Irony. Linguistic stimuli consist of short stories such that the correct understanding of the literal vs. non-literal interpretation of a target utterance depends on reasoning about the different mental states of two characters. This allows us to assess whether models actually adjust their interpretations according to the different belief contexts in the story. We evaluated our stimuli across four types of contexts and ten distinct models, including instruction-tuned ones. Our findings suggest several important points:

1. LLMs (even the best ones) perform worse than humans in most pragmatic tasks;
2. LLMs generally encounter difficulties in decoding non-literal meanings, even when these meanings are more conventionalized than their literal counterparts;
3. the model's interpretation of the target utterance appears to be independent of the belief context in the stories, indicating that even when a LLM selects the correct intended reading this is likely not to be based on a genuine ToM ability.

Our results also indicate that while instruction tuning and model size enhance the models' capabilities in performing pragmatic tasks that require reasoning about the story's characters and beliefs, instructions alone are insufficient for simulating ToM abilities. Specifically, models encounter challenges in recognizing non-literal intended meanings compared to literal meanings, often extending literal interpretations to both types of contexts.

Overall, the absence of significant differences in the results across various context setting – ToM° in which the stories contain different belief scenarios and the ToM- ones that instead contain a null or ToM+ neutral context – suggests that the LLM interpretation of utterances involving ironic expressions and indirect speech acts does not derive from inferences about the mental states of the story characters. Therefore, extreme caution should be exercised when talking of the alleged ToM abilities of LLMs, which are likely to be more apparent than real. In fact, their behavior appears to be influenced by other factors, such as inherent biases in the interpretation they tend to assign to different pragmatic constructions.

Conventionalization appears to influence not the determination of the correct belief per se, but rather the model's ability to incorporate context into the answer selection process. A more detailed investigation of the factors that shape model interpretation is left to future research, with a focus on examining the role of conventionalization in greater depth and exploring how model conventionalization differs from human conventionalization.

8 Limitations

We acknowledge several limitations in our study. While we proposed that conventionalization might affect meaning interpretation, the specific cues models use for task completion remain unclear (Ivanova, 2023), and assessing conventionalization levels is inherently subjective. Although we based our tasks on established psycholinguistic methods, the criteria for a valid Theory of Mind (ToM) test in models are still ambiguous. (Ma et al., 2023a).

Models are also highly sensitive to prompt instructions, with even minor changes leading to significantly different results (Sclar et al., 2024).

We focused on collecting metalinguistic judgments (Hu et al., 2024) rather than rely on direct measurements. Our goal was to assess the models' capacity to complete tasks that depend on ToM. We argue that probability measures of logits would be uninformative if the models do not fully grasp the task, and there is no guarantee that these probabilities pertain to the phenomena under investigation. Consequently, neither method allows us to draw definitive conclusions about the models' generalization abilities. This uncertainty is reflected in instances where models failed to select one of the proposed options, which we duly considered.

In our task, we believe the generated answers better align with real-world use cases and human judgment (Li et al., 2024) than direct measurement would. Nonetheless, this approach does not provide insights into the models' internal encodings or their mechanistic interpretability (Rai et al., 2024). However, it enables direct comparison with human performance and offers a means to assess model alignment.

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A Appendix

A.1 Stimuli and prompts

In Table 10 we present stimuli for each phenomena in our dataset. Stimuli were presented both to model and humans with slight changes. We tested humans through the Prolific platform with the following premise:

In this test, you will read short stories and you will have to answer a question about one character of the story. READ EACH STORY CAREFULLY! You must chose one among 4 possible answers. Choose the answer that you think to be correct given the story.
The test contains thee check questions. IMPORTANT. The participant will not be paid unless it answers to these questions correctly.

The check questions included in stimuli sumbitted to humans were the following:

Today is Sunday and after school Tommaso goes out to play with Red in the garden. When he hears Lisa call "Lunch is ready, time to stop playing now and come in". He runs in and sits at the table.
What does Lisa inted to say?
1 The lunch is ready; 2 The lunch isn't ready; 3 I want to go out fot lunch; 4 It's time for playing.

"Amy!" Mrs. Grove called from the door that opened towards the garden. But no answer came.
What does Mrs. Grove intend to say?
1 She is asking a question; 2 She is calling Amy; 3 She is answering a question; 4 She wants to change her name.

My boyfriend David, is a dentist. David says that he is happy to work in the centre of a small city. He can go shopping in his lunch hour, or go for a walk and look at the ducks on the river. He doesn't need a car, unless he wants to drive out of York into the countryside.
What does David intens to say?
1 He is happy to work in the centre of a small city; 2 He doesn't like working in a small city; 3 He wants to move abroa; 4 He hates the small cities.

The sequence of stories within the form, as well as the options presented in each story, were randomized. This randomization minimizes subjective error in human psycholinguistic text construction, as the entire process of form creation and stimuli randomization was automated. To generate forms from the table, we employed App Scripts. The Excel files containing human responses, along with the order of stimuli and samples, are stored on GitHub.

The query used to prompt the models, instead, was the following (example extraxted from indirect declinations):

"Task: I'll give you a story and I'll ask you to answer a question about one character of the story. I'll give you four possible answers to the question and you must choose the right one. The possible answers are 1, 2, 3 and 4. Story: I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin to pass me the glass so I can pour him more wine, but he tells me: "I need to drive". What does Kevin intend to say? Options: 1 Please, do not pour me more wine. 2 I drive tonight 3 I don't like this wine. 4 We should drink French wine. The correct answer is:"

B Appendix

B.1 ToM- and ToM+ prompts

In our second experiment, we aimed to determine whether context influenced task completion, so we designed two different prompts. The first set of stimuli consisted solely of the final remarks (irony or ISA) presented without any surrounding context (ToM-).

The second set of stimuli included a neutral context that did not favor either a literal or non-literal interpretation (ToM+).

Tables 4 and 5 present significance comparisons between ToM+ (with neutral context) and ToM° for non literal interpretations. The notation used is detailed in Section 6.

Tables 6 and 7 show comparisons between ToM- without context and ToM° for literal interpretations of irony and ISAs.

Tables 8 and 9 display comparisons between ToM+ with one neutral context and ToM° for literal interpretations of irony and ISAs.

Model	Indirect Hyperbole	Indirect Rhetoric Questions	Indirect Sarcasm
Flan-T5	†	†	-
Falcon	†	†	†
Falcon-Instruct	†	†	†
Llama2	†	†	†
Llama2-Instruct	†	†	†
GPT-4	*	†	†

Table 4: Irony (Non-literal) - Significance Comparison between ToM+ and ToM°

Model	Indirect Declinations	Indirect Requests (Ns)	Indirect Requests (Os)	Indirect Suggestions	Indirect Threats
Flan-T5	†	†	†	†	†
Falcon	†	†	†	†	†
Falcon-Instruct	†	†	†	†	†
Llama2	†	†	†	†	†
Llama2-Instruct	†	†	*	†	†
Tk-Instruct3b	†	†	†	†	†
GPT-3.5	†	†	†	†	†
GPT-4	†	†	*	†	†

Table 5: ISAs (Non-literal) - Significance Comparison between ToM+ and ToM°

Model	Indirect Hyperbole	Indirect Rhetoric Questions	Indirect Sarcasm
Flan-T5	†	†	-
Falcon	†	†	†
Falcon-Instruct	†	†	†
Llama2	†	†	†
GPT-3.5	†	*	*
GPT-4	*	*	†

Table 6: Irony (Literal) - Significance Comparison between ToM- and ToM°

Model	Indirect Declinations	Indirect Requests (Ns)	Indirect Requests (Os)	Indirect Suggestions	Indirect Threats
Flan-T5	†	†	†	†	†
Falcon	†	†	†	†	†
Falcon-Instruct	†	†	†	†	†
Llama2	†	†	†	†	†
Llama2-Instruct	†	†	†	†	†
Tk-Instruct3b	†	†	†	†	†
GPT-3.5	†	†	†	†	†
GPT-4	*	†	†	†	†

Table 7: ISA (Literal) - Significance Comparison betweem ToM- and ToM°

Model	Indirect Hyperbole	Indirect Rhetoric Questions	Indirect Sarcasm
Flan-T5	†	†	†
Falcon	†	†	†
Falcon-Instruct	†	†	†
Llama2	†	†	†
Llama2-Instruct	†	†	†
GPT-3.5	†	†	†
GPT-4	†	†	†

Table 8: Irony (Literal) - Significance Comparison between ToM+ and ToM°

Model	Indirect Declinations	Indirect Requests (Ns)	Indirect Requests (Os)	Indirect Suggestions	Indirect Threats
Flan-T5	†	†	†	†	†
Falcon	†	†	†	†	†
Falcon-Instruct	†	†	†	†	†
Llama2	†	†	†	†	†
Llama2-Instruct	†	†	†	†	†
Tk-Instruct3b	†	†	†	†	†
GPT-3.5	†	†	†	†	†
GPT-4	†	†	*	†	†

Table 9: ISAs (Literal) - Significance Comparison between ToM+ and ToM°

Phenomenon	Literal Story	Non-Literal Story	Question	Options
Indirect Request (Os)	You and your friend Jonathan are taking a road trip. You began in California, and are now passing through Michigan. It's almost winter, so it's very cold outside-especially for Southern California dwellers like you and Jonathan. You see that you're almost out of gas, so you stop at a gas station in a small town. You fill up the tank, and then the two of you go inside the gas station to buy some water and snacks. When you return to the car and start up the engine, you and Jonathan notice with some dismay a blinking light, which indicates that the car's heating system is broken. You both bundle up. As you leave the station, Jonathan shivers in his seat. He turns to you and says, "Man, it's really cold in here"	You and your friend Jonathan are taking a road trip. You began in California, and are now passing through Michigan. It's almost winter, so it's very cold outside-especially for Southern California dwellers like you and Jonathan. You see that you're almost out of gas, so you stop at a gas station in a small town. While you fill up the tank, Jonathan goes inside to buy some water and snacks. As you're checking the meter, you notice with some dismay a blinking light, which indicates that the car's heating system is broken. You finish filling up the gas and wait for Jonathan. Jonathan returns with some snacks, and you both set off. As you leave the station, he shivers in his seat. He turns to you and says, "Man, it's really cold in here."	What does Jonathan intend to say?	1. Could you turn on the heater? 2. I'm really cold; it's too bad the heater is broken. 3. We should fix the heater now. 4. We should get out of the car to warm up.
Indirect Suggestions	Noah lives in New Jersey, but his girlfriend Rachel lives in California. Noah often complains to his friend Jordan that he never gets to spend time with Rachel and that he misses her a lot. Today, Jordan and Noah are out to lunch and Noah is talking about how he decided to quit his job and move to Florida. Jordan is surprised by this decision and says: "Why don't you move to California?"	Noah lives in New Jersey, but his girlfriend Rachel lives in California. Noah often complains to his friend Jordan that he never gets to spend time with Rachel and that he misses her a lot. Today, Jordan and Noah are out to lunch and Noah is complaining about his long-distance relationship. Jordan knows that Noah would really like to move in with Rachel and that he isn't fully satisfied with his life in New Jersey. Jordan at one point says: "Why don't you move to California?".	What does Jordan intend to say?	1. I suggest you move to California. 2. Why are you moving to Florida and not California? 3. You shouldn't move to California. 4. California is uglier than Florida.
Indirect Declinations	I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin why he stopped drinking. Kevin replies "I need to drive".	I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin to pass me the glass so I can pour him more wine, but he tells me: "I need to drive".	What does Kevin intend to say?	1. Please, do not pour me more wine. 2. I drive tonight. 3. I don't like this wine. 4. We should drink French wine.
Indirect Threats	Janet and Monica are planning to go on a trip to the lake. They decide to invite their friend Brittany too. However, Brittany decides to join them after the picnic at Janet's house. Janet asks if she wants them to pick her up before they go home, but Brittany replies: "I know where you live".	Janet and Monica are planning to go on a trip to the lake. They decide to invite their friend Brittany too. However, Brittany decides to join them after the picnic at Janet's house. Once at Janet's house, Brittany reminds Janet that she has to pay her back the money she lent her and adds "I know where you live."	What does Brittany intend to say?	1. Give me my money back or I will come looking for you. 2. I know where your house is. 3. I really like your house. 4. I don't want to come to your house.
Sarcasm	Kyle and Mark play soccer on a team in their town. Today their team is playing the final. Kyle and Mark both play offense. During the match Mark makes a pass to Kyle and Kyle scores a goal. Mark tells him "Nice shot."	Kyle and Mark play soccer on a team in their town. Today their team is playing the final. Kyle and Mark both play offense. During the match Mark makes a pass to Kyle but Kyle misses the shot and fails to score a goal. Mark tells him "Nice shot.	What does Mark intend to say?	1. It was a really bad shot. 2. It was a really good shot. 3. You should try again. 4. You should take a shot.
Indirect Hyperbole	Monica and Rachel are two old friends who haven't met since university. After speaking by phone, they decide to meet for a coffee to see each other again and update on their lives. When they meet, Monica says "Haven't seen you in ages."	Monica and Rachel are two best friends. They went to the same high school and the same university, live in the same house, and practically spend all their time together. Tonight, Monica decides to go out with some of her friends and while she is at a restaurant she meets Rachel who is having dinner with her boyfriend. Monica tells Rachel "Haven't seen you in ages."	What does Monica intend to say?	1. We always see each other. 2. Haven't seen each other for a long time. 3. I don't want to see you again. 4. I would like to see you again soon.
Rhetorical Questions	Clara attends the art academy and often during the courses they must copy famous paintings. During today's lesson, Clara must reproduce a painting of her teacher. Clara is sitting in the last row and cannot distinguish the colors of the painting well, also because the colors are very dark. After drawing the draft, she begins to color, but is undecided about which are the right tempera colors to use, so she says to the teacher "Is the sky blue?"	Clara attends the art academy and often during the courses they must copy famous paintings. During today's lesson, Clara must reproduce a painting of her teacher. Clara is sitting in the last row and cannot distinguish the colors of the painting well, also because the colors are very dark. After drawing the draft, she begins to color. At a certain point one of her classmates asks Clara if she will use the brush to color and Clara says, "Is the sky blue?"	What does Clara intend to say?	1. The answer is obvious. 2. I want to know if the color of the sky is blue. 3. I think the sky is blue. 4. I don't think the sky is blue.

Table 10: Stimuli for each phenomenon, illustrating the two versions of the story. Literal correct answers are with number 2 and non-literal correct answers with number 1.